

Wisdom Science Physics and Engineering Core: State Dynamics, Stability, Cost, and Acceptance Tests

Mian Zhang

Abstract

Wisdom Science should not rely on vague metaphors. If it borrows language from mathematics, physics, and engineering, the borrowed terms must become measurable objects. This document gives the physics and engineering core of the program. We define agent state dynamics, plasticity as normalized response to experience, homeostasis as bounded perturbation response, failure immunity as repeat-failure attenuation, and wisdom efficiency as improvement per resource budget. We also specify release gates: a result is not publishable unless it passes metric, stability, provenance, cost, and duplicate-cell checks.

1 No Empty Metaphor Rule

A term is admissible in Wisdom Science only if it satisfies three conditions:

1. It names a measurable quantity or an explicit hypothesis.
2. It has a failure condition under which the claim is rejected.
3. It can be connected to a script, table, raw log, or proof obligation.

This rule prevents the framework from becoming decorative philosophy. Physics language is useful only when it constrains measurement. Engineering language is useful only when it constrains deployment.

2 State Dynamics

Let $x_r \in \mathcal{X}$ be an agent state at round r : policy parameters, memory state, context, retrieved evidence, failure antibodies, and controller state. Let u_r be the task input, e_r the experience or feedback signal, and ξ_r a perturbation. The agent evolves as

$$x_{r+1} = F_{\mathcal{A}}(x_r, u_r, e_r, \xi_r),$$

where $\mathcal{A}$ denotes architecture. A score functional $S_t(x_r) \in [0, 1]$ maps state to task performance. Wisdom Science studies whether repeated application of $F_{\mathcal{A}}$ moves the system toward better, more stable, and more transferable behavior.

Definition 1 (Plasticity). *Plasticity is normalized response to useful experience:*

$$\Phi = \mathbb{E}_{t,r} \left[\frac{\max(0, S_t(x_{r+1}) - S_t(x_r))}{\|e_r\| + \epsilon} \right],$$

reported with the experience representation and scoring rule.

Definition 2 (Homeostasis). *Homeostasis is inverse perturbation gain:*

$$H = \frac{1}{1 + \mathbb{E}_{t,r,\xi} [|S_t(F(x_r, u_r, e_r, \xi)) - S_t(F(x_r, u_r, e_r, 0))|]}.$$

High H means that perturbations produce small score deviations.

Definition 3 (Failure immunity). *For a failure family a , let $p_1(a)$ be repeat-failure probability before an antibody or memory update, and $p_R(a)$ after repeated experience. Define*

$$\Psi(a) = 1 - \frac{p_R(a)}{p_1(a) + \epsilon}.$$

Transferable immunity requires $\Psi(a)$ to be larger on related tasks than unrelated controls.

Definition 4 (Wisdom efficiency). *Let ΔW be a measured wisdom gain and C be normalized resource cost including GPU time, wall time, API spend, storage, or robot wear.*

$$\eta_W = \frac{\Delta W}{C + \epsilon}.$$

A claim that improves score while hiding cost is incomplete.

3 Physics-Aligned Constraints

Proposition 1 (Bounded perturbation response). *Suppose a perturbation deviation follows $d_{r+1} \leq Ld_r + \epsilon$ with $0 \leq L < 1$. Then*

$$d_r \leq L^r d_0 + \frac{\epsilon(1 - L^r)}{1 - L}.$$

Thus a contractive homeostatic response keeps deviation bounded.

Proof. Iterating the recurrence gives $d_r \leq L^r d_0 + \epsilon \sum_{k=0}^{r-1} L^k$. The geometric sum equals $(1 - L^r)/(1 - L)$. $\square$

Proposition 2 (Plasticity without homeostasis can be fragile). *An update rule can have large response gain Φ and still reduce deployable wisdom if perturbation amplification or overshoot increases repeat failures.*

Proof. Let an update improve nominal score by Δ but amplify perturbation loss by $\Delta + \delta$ on the deployment distribution. The net deployed score change is $-\delta$. Therefore high plasticity alone is insufficient; homeostasis and failure immunity must be measured jointly. $\square$

Remark 1. *This is the physics meaning of the Second Scaling Law: wisdom is not only acceleration. It is adaptive response constrained by stability, transfer, and resource budgets.*

4 Engineering Contract

A Wisdom Science result should be shipped as an engineering contract:

Contract field	Required content
Task contract	task IDs, strata, scoring rule, task weights
Round contract	number of rounds, feedback visibility, reset policy
Policy contract	model, checkpoint, factory, adapter, version hash
Evidence contract	raw logs, seeds, metadata, trajectory or transcript IDs
Metric contract	I , WQ or I_E , EWQ, Φ , Ψ , H , cost
Stress contract	perturbations, held-out tasks, duplicate-cell checks
Release contract	admissible claims, limitations, failure conditions

Table 1: Engineering contract for non-empty Wisdom Science claims.

Table 2: Physics and engineering executable checks.

Check	Pass	Engineering meaning
Dimensionless metrics	yes	Physics-inspired variables are normalized measurable ratios, not decorative symbols.
Homeostasis damping	yes	Higher H must reduce integrated perturbation deviation in the toy response model.
Plasticity-stability tradeoff	yes	Plasticity without homeostasis can underperform a balanced adaptive regime.
Failure transfer	yes	A valid antibody should attenuate repeat failures in related tasks more than unrelated tasks.
Cost budget	yes	Engineering claims must report compute/cost budgets, not only performance deltas.
Release gate	yes	A publishable row must pass metric, stability, provenance, budget, and duplicate-cell gates.
Falsification gate	yes	A positive delta is rejected when checkpoint provenance is missing or the task mixture changed.

5 Acceptance Tests

The following acceptance tests are required before a leaderboard row, robot result, or public-checkpoint result is treated as strong evidence:

1. The score metric is normalized and the task mixture is fixed or stratified.
2. WQ/EWQ is accompanied by raw first-round and final-round scores.
3. Positive WQ/EWQ is not interpreted as learning unless provenance supports the mechanism.
4. Perturbation response is reported through H or an equivalent stability statistic.
5. Failure transfer is reported through related and unrelated controls.
6. Resource cost is reported.
7. Raw logs, seeds, metadata, checkpoint/factory identity, and hashes are present.

6 Executable Checks

The package includes dependency-free checks that instantiate the constraints above. They are not substitutes for large experiments. They are engineering tripwires: if the toy checks fail, the language in the papers is already too loose.

7 What Would Falsify a Claim

A Wisdom Science claim is weakened or rejected under any of the following conditions:

- The task mixture changes across compared systems without stratified reporting.
- The result has positive WQ/EWQ but no checkpoint, memory-state, trajectory, or feedback provenance.
- The reported improvement disappears under perturbation.
- The same raw trajectory or checkpoint is duplicated under multiple policy names.
- The gain is smaller than the cost of the representation, memory, adapter, or compute budget.
- The result cannot be regenerated from scripts and manifests.

8 Conclusion

The physics and engineering core turns Wisdom Science from a slogan into a testable system. Plasticity is a response coefficient. Homeostasis is perturbation damping. Failure immunity is repeat-failure attenuation with transfer controls. Wisdom efficiency is gain per resource. Evidence gates are release gates. This framing keeps the program bold without letting it become empty.